

CENTRAL AI TECHNICAL ASSESSMENT

House Prices: Advanced Regression Techniques | Technical Report

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1. Executive Summary & Problem Formulation

This report documents an end-to-end machine learning solution for the Ames Housing price prediction task (Kaggle: House Prices – Advanced Regression Techniques). The objective is to predict residential property sale prices using 79 explanatory features describing structural attributes, spatial configurations, materials, and condition metrics.

The evaluation metric specified is Root Mean Squared Logarithmic Error (RMSLE). A structured, leak-free machine learning pipeline was engineered, incorporating domain-driven feature synthesis, rigorous 5-fold cross-validation, and an ensemble blending of regularized linear models with gradient boosted decision trees. Our final blended ensemble achieved a 5-fold cross-validation RMSLE of 0.1126 ($R^2 = 0.920$), demonstrating superior generalization.

2. Exploratory Data Analysis & Empirical Discoveries

Exploratory analysis on the 1,460 training and 1,459 testing observations revealed several critical data properties:

- **Target Skewness:** Raw SalePrice exhibited pronounced right-skewness (skew = 1.88, kurtosis = 6.54). Applying the logarithmic transformation $\tilde{y} = \log(1 + \text{SalePrice})$ normalized the distribution (skew = 0.12, kurtosis = 0.81), perfectly aligning model optimization with the competition RMSLE metric.
- **Missing Value Taxonomy:** Missingness in Ames housing is primarily structural rather than stochastic. Out of 34 features with missing data, 15 categorical features (e.g., PoolQC, MiscFeature, Alley, FireplaceQu, GarageType, BsmtQual) use 'NA' to indicate the physical absence of an amenity. Treating these as random dropouts would corrupt signal; they must be imputed as 'None' (and 0 for numerical counterparts).
- **Primary Linear Drivers:** The strongest correlates with $\log(\text{SalePrice})$ are OverallQual ($r = 0.817$), GrLivArea ($r = 0.701$), GarageCars ($r = 0.681$), and TotalBsmtSF ($r = 0.612$).
- **Severe Outlier Verification:** Dataset creator Dean De Cock explicitly advises removing homes with GrLivArea > 4,000 sq ft. In our training set, Id 524 (4,676 sq ft) and Id 1299 (5,642 sq ft) represent partial/abnormal sales sold far below market value. Both were safely pruned prior to model training to prevent distorted split gradients.

3. Data Preprocessing & Domain Feature Engineering

To maximize predictive signal without data leakage, all statistical parameters were computed strictly on training folds:

- **Leakage-Free Imputation:** Structural categories were filled with 'None' and structural numericals with 0. LotFrontage (17.7% missing) was imputed using the median of the corresponding Neighborhood, learned strictly from training records.
- **Composite Spatial Area (TotalSF):** Created $TotalSF = TotalBsmtSF + 1stFlrSF + 2ndFlrSF$. This engineered feature achieved an outstanding Pearson correlation of $r = 0.825$ with $\log(SalePrice)$, surpassing every single raw predictor in the dataset.
- **Cumulative Amenities & Temporal Metrics:** $TotalBath = FullBath + 0.5 * HalfBath + BsmtFullBath + 0.5 * BsmtHalfBath$, $r = 0.677$, $HouseAge = YrSold - YearBuilt$ ($r = -0.588$), $RemodAge$, and $OverallGrade = OverallQual * OverallCond$ ($r = 0.608$).
- **Encoding Strategy:** Natural ordinal scales (ExterQual, KitchenQual, BsmtQual, GarageFinish, etc.) were mapped to numerical ranks (0 to 5). Remaining nominal variables were one-hot encoded, and test feature columns were strictly aligned with zero-fill (resulting in 227 aligned numeric predictors).

4. Validation Methodology & Model Experimentation

To prevent data leakage and provide reliable generalization estimates, we implemented 5-Fold Cross-Validation with shuffle (random_state=42). Five diverse machine learning algorithms representing regularized linear models, bagging, and sequential boosting were benchmarked:

Model Architecture	Paradigm	5-Fold CV RMSLE	Std Dev	OOF R ²
Ridge Regression	L2 Linear Shrinkage	0.1146	± 0.0079	0.916
Lasso Regression	L1 Feature Sparsity	0.1149	± 0.0073	0.915
Gradient Boosting (GBR)	Sequential Boosting	0.1191	± 0.0074	0.910
LightGBM Regressor	Leaf-Wise Boosted Trees	0.1220	± 0.0084	0.906
Random Forest Regressor	Bagging Ensemble	0.1338	± 0.0074	0.887

Winning Blended Ensemble Strategy

Because linear regularized models excel in high-dimensional sparse spaces and gradient boosted trees excel at non-linear interactions, their prediction errors are largely orthogonal. Combining predictions using an optimal weighted blend:

$$\hat{y}_{\text{ensemble}} = 0.40 \cdot \hat{y}_{\text{Ridge}} + 0.30 \cdot \hat{y}_{\text{LightGBM}} + 0.30 \cdot \hat{y}_{\text{GBR}}$$

reduced out-of-fold RMSLE to 0.1126 ($R^2 = 0.920$), outperforming every individual standalone model.

5. Error Analysis & Model Diagnostics

A thorough analysis of model residuals ($e_i = y_i - \hat{y}_i$) demonstrated strong statistical validity:

- **Residual Normality:** Residual distribution is centered at zero (mean = -0.0003, std = 0.1126) with near-Gaussian bell shape. Normal Q-Q probability plots demonstrate tight linear alignment with minor tail deviations.
- **Homoscedasticity:** Scatter plots of residuals versus fitted values confirm constant variance across price tiers (\$50k to \$750k), verifying that logarithmic scaling eliminated price-dependent heteroscedasticity.
- **Failure Mode Investigation:** Properties yielding the largest residuals corresponded to rare atypical records, such as severe structural zoning restrictions (commercial zoning 'C (all)') and unique abnormal distress sales.
- **Feature Importances:** In LightGBM, the engineered metric TotalSF ranked as the #1 most important feature by split count, followed by OverallQual, OverallGrade, and HouseAge, confirming that domain synthesis directly empowered tree partitioning.

6. Final Submission & Repository Architecture

The final blended ensemble was refit on the full 1,458 preprocessed training observations and applied to generate predictions for all 1,459 competition test properties. Predictions were inverted via $\exp(\hat{y}) - 1$ and validated against ground-truth distributions:

- Mean Predicted Price: \$178,241.87 (vs. \$180,921.20 in training data)
- Median Predicted Price: \$155,712.52 (vs. \$163,000.00 in training data)
- Prediction Range: \$50,838.02 to \$970,088.14 (0 nulls, 0 non-positive values)

The competition-ready submission was exported to outputs/submission.csv.